Fangyi Yu

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Education

New York University *Sep.2025~Present*

M.S. in Electrical Engineering GPA: 3.762/4.0

- Core Courses: Digital Signal Processing (A), Computer Systems Architecture (A-), Introduction to Machine Learning (B), Design and Analysis of RFIC (A-), LLM4ChipDesign (A)

Lanzhou University *Sep.2021~Jun 2025*

B.S. in Microelectronics Science and Engineering GPA:84.96/100

- Core Courses: Quantum Mechanics (94), Method of Mathematical Physics (97), Solid-State Electronic Device (93), Physics of Semiconductors (89), Microelectronics Manufacturing Technology (87)

Research Experience

Intent-Aware LLMs for Conversational Timing Constraint Generation

Advised by Prof. Austin Rovinski, New York University *Sep 2026 – Present*

- Built a two-tier benchmark for timing constraint generation in Synopsys Design Constraints (SDC) format, with 15 basic tasks over seven scenarios and 10 tasks on realistic designs such as FIFOs
- Designed experiments comparing direct generation with interactive clarification under incomplete constraint specifications, across three prompt levels of design information
- Implemented syntax, semantic, and timing-report checks against reference SDC files and synthesized netlists in OpenROAD to score generated constraints
- Benchmarked four open-source and three closed-source LLMs with error attribution across syntax errors, missing constraints, and tool-call errors
- Currently expanding the seed dataset to a scale suitable for LoRA fine-tuning to improve SDC generation accuracy

Deep Research Benchmark for EDA

With PhD students Zeng Wang and Minghao Shao, New York University *Jul 2026 – Present*

- Co-developing a deep-research benchmark that evaluates LLM agents on open-ended EDA research tasks
- Built a task set of 100+ evaluation questions across 14 research topics organized along the EDA flow
- Currently refining question content ahead of evaluating deep-research agents on the benchmark

Recurrent-State Control of Retrieval in Hybrid Vision-Language Models

Team project, New York University *Oct 2026 – Present*

- Led the prompt-injection evaluation, benchmarking the proposed recurrent-state write defense against DataFilter, Attention Tracker, and Spotlighting on 512 tasks across two hybrid VLMs
- Designed and ran the white-box adaptive attack (GCG) evaluation on the 9B model, adding two new attack conditions and cross-model transfer tests
- Traced the defense's failure on a multi-turn tool-calling agent (InjecAgent, 1,054 cases) to the recurrent state tracking task progress; skip writes restored utility from 15.7% to 83.6% at 1.5% ASR
- Designed KV/state swap experiments on CV-Bench, TallyQA, and What'sUp, showing that whether answers reside in the recurrent state or KV cache depends on answer format, difficulty, and layout

Diffusion-Based Speculative Decoding for Efficient LLM Inference

Advised by Prof. Bo Yuan, Rutgers University *Mar 2026 – Present*

- Applied Mosaic's chunked logits computation to long-context UltraLLaDA-8B inference, reducing peak GPU memory by ~40% and enabling 128K-context inference on a single GPU
- Characterized weight-only quantization sensitivity of the DFlash drafter: INT8/INT4 keep acceptance length within 1% of BF16, while INT2 cuts it by about one third
- Proposed a training-free relaxed verification rule that uses DSpark's confidence head to accept mismatched draft tokens, evaluated on GSM8K against Approximate Speculative Decoding
- Proposed a second rule that uses DSpark's Markov head to estimate how replacing a mismatched token shifts the next-position prediction, turning it into an acceptance criterion

Research on the Sensor Performance Based on WS_2

Advised by Prof. Yanrong Wang, Lanzhou University *Mar – Jun 2025*

WS_2-based gas sensors with high sensitivity to ethanol for monitoring ethanol vapor in public environments, contributing to the prevention of fire hazards and gas poisoning

- Prepared high-purity tungsten disulfide (WS_2) using a controlled hydrothermal synthesis process for the fabrication and optimization of a high-performance gas sensor
- Characterized the crystal structure and morphology of WS_2 material using scanning electron microscopy (SEM) and transmission electron microscopy (TEM)
- Conducted performance testing for the fabricated gas sensor including testing its optimal operating temperature, selectivity for different gases, and gradient performance for ethanol gas
- Analyzed sensor response characteristics under varying operating conditions, including response curves, sensitivity, and stability with respect to temperature and gas concentration

Selected Projects

2.4-GHz Direct-Conversion CMOS Receiver Signal-Path Design

Course project, Prof. Hamed Rahmani, New York University *Apr – May 2026*

Architected a full-chain RF front-end (LNA, Gilbert-cell Mixer, IF Stage) under stringent 100 nH total on-chip inductance and 1.8V supply constraints

- Designed and simulated the RF-to-baseband signal path of a 2.4-GHz direct-conversion CMOS receiver, integrating an inductively degenerated cascode LNA, double-balanced Gilbert-cell mixer, and differential IF gain stage
- Optimized LNA input matching and gain through transistor sizing, bias tuning, and inductor/capacitor sweeps, achieving low input return loss, high forward gain, and reducing standalone noise figure across the 2.40–2.48 GHz band
- Built and verified the complete receiver signal-path schematic in Cadence Virtuoso, using SpectreRF simulations for DC biasing, S-parameter matching, noise figure, conversion gain, bandwidth, and two-tone IIP3 characterization
- Balanced gain, noise, linearity, impedance matching, and passive-component constraints, achieving 37.1 dB receiver gain, 50 Ω input matching, 700 Ω output resistance, and compliance with the 100 nH total on-chip inductance limit

AI-based Hardware Attacks (AHA!) Challenge

Course project, Prof. Ramesh Karri, New York University *Apr 2026*

Designed and evaluated LLM-driven hardware Trojans in a red-team setting to advance understanding of AI-enabled hardware security vulnerabilities and attack methodologies.

- Conducted a vulnerability assessment of the Secworks AES core by integrating LLM-generated hardware Trojans, optimized its evasive stealth and minimal hardware footprint to bypass RTL-level detection
- Trojan 1: Elicited a functional-logic Trojan on the AES core's control path from an LLM in a single prompt; it stays dormant during normal encryption, keeping power and area overhead low
- Trojan 2: Produced a "time bomb" Trojan through a few rounds of LLM prompting; its 8-bit counter fires after 100 valid encryptions and leaks the lower 32 bits of the secret key
- Trojan 3: Built a denial-of-service Trojan through multi-turn LLM interaction; it detects a three-stage input sequence and locks the control FSM in a busy state (ready = 0) until reset

Publications

- Srinivas Harish, **Fangyi Yu**, Anshul Singh, Sai Qian Zhang. "Recognized in the State, Read from the Cache: The Recurrent State Controls What Hybrid Vision-Language Models Retrieve." *Manuscript in preparation, targeting CVPR 2027.*
- **Fangyi Yu.** "Research on the Development of LDO Chip Technology." *The 2nd International Conference on Machine Learning and Automation*, 2024.
- **Fangyi Yu.** "Research and Analysis on Artificial Intelligence in Integrated Circuits." *2nd International Conference on Artificial Intelligence, Database and Machine Learning (AIDML 2024)*, 2024.

Skills

- **Languages:** Mandarin (native), English (fluent; IELTS 7.0)
- **Programming:** Python, Verilog, Tcl
- **ML & LLM Tools:** PyTorch, Hugging Face Transformers, torchao, LoRA
- **EDA & Circuit Design:** OpenROAD, Yosys, OpenSTA, Cadence Virtuoso, SpectreRF
- **Other:** Git, Linux, MATLAB